Technical Note: block-DDA_Py

Discrete Dipole Approximation with FFT-accelerated
Block-Krylov Solver for Light Scattering by Arbitrary Particles

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Contents

1 Introduction

The discrete dipole approximation (DDA) is a flexible numerical method for computing electromagnetic scattering and absorption by particles of arbitrary shape and composition [1, 2]. The target particle is represented by an array of N point dipoles on a cubic lattice with spacing d. Each dipole acquires a polarisation in response to the local electric field, which consists of the incident wave plus the fields radiated by all other dipoles.

This document describes the theoretical background and the algorithms that are implemented in version 0.4.0 of `block-DDA_Py`, covering:

1. the DDA formulation and polarisability prescription (Sec. 2),
2. the Goodman (1991) FFT-accelerated matrix–vector product (Sec. 3),
3. the block-Krylov iterative solvers (Sec. 4),
4. the scattering observables computed from the solution (Sec. 6),
5. the Gaussian Random Ellipsoid (GRE) shape model (Sec. 7),
6. the Mie theory reference solution for homogeneous spheres (Sec. 9).

Throughout this note, the electromagnetic equations are written in Gaussian (CGS) units; lengths are expressed in micrometers (µm). Vectors are set in bold italic ($\boldsymbol{E}$, $\boldsymbol{k}$, $\boldsymbol{r}$), matrices and second-rank tensors in upright bold ($\mathbf{A}$, $\boldsymbol{\alpha}$), and unit vectors carry a hat ($\hat{\boldsymbol{e}}$, $\hat{\boldsymbol{r}}$).

2 DDA Formulation

2.1 Fundamental equation

Consider a monochromatic plane wave of vacuum wavelength λ_0 incident on a particle embedded in a non-absorbing medium of refractive index m_m. The wavenumber in the medium is

$$k = \frac{2\pi\, m_m}{\lambda_0}. \tag{1}$$

The particle is discretised into N cubic elements (dipoles) of side length d and volume $v = d^3$, placed on a regular lattice with grid dimensions $N_x \times N_y \times N_z$. The lattice position of the i-th dipole is $\boldsymbol{r}_i$.

The polarisation $\boldsymbol{P}_i$ of dipole i satisfies

$$\boldsymbol{P}_i = \boldsymbol{\alpha}_i \cdot \boldsymbol{E}(\boldsymbol{r}_i), \tag{2}$$

where $\boldsymbol{\alpha}_i$ is the 3×3 polarisability tensor and $\boldsymbol{E}(\boldsymbol{r}_i)$ is the total electric field at position $\boldsymbol{r}_i$. The total field is the superposition of the incident field and the contributions from all other dipoles:

$$\boldsymbol{E}(\boldsymbol{r}_i) = \boldsymbol{E}_{\mathrm{inc},i} - \sum_{j \neq i} \mathbf{G}(\boldsymbol{r}_i - \boldsymbol{r}_j) \cdot \boldsymbol{P}_j, \tag{3}$$

where $\mathbf{G}$ is the free-space dyadic Green's function (Sec. 2.3). Substituting (3) into (2) and rearranging yields the $3N \times 3N$ linear system

$$\mathbf{A}\,\boldsymbol{P} = \boldsymbol{E}_{\mathrm{inc}}, \tag{4}$$

where $\boldsymbol{P}$ and $\boldsymbol{E}_{\mathrm{inc}}$ are column vectors of length $3N$ formed by concatenating the three Cartesian components of each dipole (ordering: $P_{x,0}, P_{y,0}, P_{z,0}, P_{x,1}, \ldots$), and $\mathbf{A}$ is a complex-valued matrix with

$$\mathbf{A}_{ij} = \begin{cases} \boldsymbol{\alpha}_i^{-1} & \text{if } i = j, \\ \mathbf{G}(\boldsymbol{r}_i - \boldsymbol{r}_j) & \text{if } i \neq j. \end{cases} \tag{5}$$

When the dipoles are placed on a regular cubic lattice, the off-diagonal blocks $\mathbf{A}_{ij}$ depend only on the difference $\boldsymbol{r}_i - \boldsymbol{r}_j$, giving $\mathbf{A}$ a block-Toeplitz structure that can be exploited by FFT (Sec. 3).

2.2 Incident field

The incident plane wave in the medium is

$$\boldsymbol{E}_{\mathrm{inc},i} = \boldsymbol{E}_0 \exp(i\boldsymbol{k} \cdot \boldsymbol{r}_i - i\omega t), \tag{6}$$

where $\boldsymbol{k} = k\,\hat{\boldsymbol{u}}_{\mathrm{inc}}$ with $\hat{\boldsymbol{u}}_{\mathrm{inc}}$ the propagation direction (fixed to $+z$ in the laboratory frame). The time-harmonic factor $e^{-i\omega t}$ is suppressed hereafter.

To simulate orientation-averaged scattering, the particle is rotated through $L = N_\alpha \times N_\beta \times N_\gamma$ orientations on a deterministic grid of intrinsic ZYZ Euler angles:

- α: N_α equally spaced values in $[0, 2\pi)$,
- β: N_β values chosen so that $\cos\beta$ is equally spaced in $(-1, 1)$ (equal-area midpoints on the sphere),
- γ: N_γ equally spaced values in $[0, 2\pi)$.

This deterministic scheme avoids sampling noise and ensures reproducibility. For each orientation, the laboratory-frame vectors are transformed into the particle frame via the inverse rotation matrix. Batch rotation is performed using `scipy.spatial.transform.Rotation`.

Following the CAS-v2 (Complex Amplitude Sensing, version 2) beam configuration, the incident polarisation is assumed to be left-handed circular. This setting is hard-coded in the current version (v0.6.0). The polarisation vector is:

$$\hat{\boldsymbol{e}}_0 = \frac{1}{\sqrt{2}}\,\hat{\boldsymbol{e}}_\theta + \frac{i}{\sqrt{2}}\,\hat{\boldsymbol{e}}_\phi, \tag{7}$$

where $\hat{\boldsymbol{e}}_\theta$ and $\hat{\boldsymbol{e}}_\phi$ are the polar and azimuthal unit vectors associated with the incident direction.

2.3 Free-space dyadic Green's function

The 3×3 interaction tensor for two dipoles separated by $\boldsymbol{r} = \boldsymbol{r}_i - \boldsymbol{r}_j$ ($r = |\boldsymbol{r}|$, $\hat{\boldsymbol{r}} = \boldsymbol{r}/r$) is [3]

$$\mathbf{G}(\boldsymbol{r}) = \frac{e^{ikr}}{r}\left[k^2\big(\hat{\boldsymbol{r}} \otimes \hat{\boldsymbol{r}} - \mathbf{I}_3\big) + \frac{1 - ikr}{r^2}\big(\mathbf{I}_3 - 3\,\hat{\boldsymbol{r}} \otimes \hat{\boldsymbol{r}}\big)\right], \tag{8}$$

where $\otimes$ denotes the outer (tensor) product and $\mathbf{I}_3$ the 3×3 identity matrix. The self-interaction ($\boldsymbol{r} = \mathbf{0}$) is excluded from $\mathbf{G}$ and handled separately via the diagonal $\boldsymbol{\alpha}_i^{-1}$ in (5).

2.4 Polarisability: Clausius–Mossotti with radiative correction

The relative permittivity of the i-th dipole along Cartesian axis c ($c \in \{x, y, z\}$) is

$$\epsilon_{r,c} = \frac{m_{p,c}^2}{m_m^2}, \tag{9}$$

where $m_{p,c}$ is the (generally complex) particle refractive index along axis c.

The static Clausius–Mossotti (CM) polarisability is

$$\alpha_{0,c} = \frac{3}{4\pi}\,\frac{\epsilon_{r,c} - 1}{\epsilon_{r,c} + 2}\,v, \tag{10}$$

where $v = d^3$ is the dipole volume.

To improve accuracy, the CM polarisability is corrected for radiative reaction following Chaumet & Rahmani [4]. Define the equivalent-volume sphere radius

$$a = \left(\frac{3v}{4\pi}\right)^{1/3} \tag{11}$$

and the correction term

$$M = \frac{8\pi}{3}\big[(1 - ika)\, e^{ika} - 1\big]. \tag{12}$$

The dynamical (radiative-reaction corrected) polarisability is then

$$\alpha_c = \frac{\alpha_{0,c}}{1 - M\,\alpha_{0,c}\,/\,v}. \tag{13}$$

In general the particle may be optically anisotropic, so $\boldsymbol{\alpha}_i = \mathrm{diag}(\alpha_x, \alpha_y, \alpha_z)$. For isotropic particles, $m_{p,x} = m_{p,y} = m_{p,z}$ and the polarisability reduces to a scalar times $\mathbf{I}_3$.

3 FFT-Accelerated Matrix–Vector Product (Goodman 1991)

Direct evaluation of the matrix–vector product (MVP) $\mathbf{A}\,\boldsymbol{P}$ requires $O(N^2)$ operations and $O(N^2)$ storage. Because the off-diagonal part of $\mathbf{A}$ has block-Toeplitz structure on a regular cubic lattice, the MVP can be evaluated via 3D FFT in $O(N \log N)$ time with $O(N)$ storage [3].

3.1 Interaction tensor on the doubled grid

The physical cuboid lattice has dimensions $N_x \times N_y \times N_z$. The index difference along each axis α ranges from $-(N_\alpha - 1)$ to $+(N_\alpha - 1)$. To evaluate the linear convolution as a cyclic convolution via FFT, the lattice is extended to $2N_x \times 2N_y \times 2N_z$ with the following index mapping.

For each extended-grid index $I_\alpha \in \{0, 1, \ldots, 2N_\alpha - 1\}$, the physical integer displacement $\Delta\alpha$ is

$$\Delta\alpha(I_\alpha) = \begin{cases} I_\alpha & 0 \le I_\alpha \le N_\alpha - 1 \quad \text{(positive displacements)}, \\ 0 & I_\alpha = N_\alpha \quad \text{(Nyquist; zeroed out)}, \\ I_\alpha - 2N_\alpha & N_\alpha + 1 \le I_\alpha \le 2N_\alpha - 1 \quad \text{(negative displacements)}. \end{cases} \tag{14}$$

The physical displacement vector at extended-grid index (I_x, I_y, I_z) is $\boldsymbol{r} = d\,(\Delta x,\, \Delta y,\, \Delta z)$, and the interaction tensor $\mathbf{G}(\boldsymbol{r})$ is computed via (8). Three special cases are set to zero:

- self-interaction at the origin: $\mathbf{G}(\mathbf{0}) = \mathbf{0}$,
- Nyquist positions ($I_\alpha = N_\alpha$ for any α): physically unreachable displacements.

The resulting array has shape $(2N_x, 2N_y, 2N_z, 3, 3)$ and dtype `complex128`.

3.2 Pre-computation of the FFT

Before the iterative solver begins, the 3D discrete Fourier transform of $\mathbf{G}$ along the spatial axes is computed once:

$$\widetilde{\mathbf{G}} = \mathrm{FFT}_{3\mathrm{D}}\big[\mathbf{G}\big]. \tag{15}$$

$\widetilde{\mathbf{G}}$ has the same shape $(2N_x, 2N_y, 2N_z, 3, 3)$ and is stored in memory throughout the simulation. It is independent of the incident direction, polarisation, and Krylov iteration, so this computation is performed only once per (λ_0, m_m) pair.

3.3 Block MVP via FFT convolution

Given a block of L polarisation vectors stacked as columns in $\mathbf{P}_{\text{block}}$ (shape $3N_xN_yN_z \times L$), the off-diagonal MVP $\mathbf{A}_{\text{off}}\,\mathbf{P}_{\text{block}}$ is computed as follows.

1. **Reshape** $\mathbf{P}_{\text{block}}$ from $(3N_xN_yN_z,\ L)$ to the 5D physical representation $(N_x, N_y, N_z, 3, L)$.
2. **Zero-pad** to the doubled grid:

$$\hat{\mathbf{P}}(I_x, I_y, I_z, c, \ell) = \begin{cases} \mathbf{P}(I_x, I_y, I_z, c, \ell) & \text{if } I_\alpha < N_\alpha \ \forall \alpha, \\ 0 & \text{otherwise.} \end{cases} \tag{16}$$

3. **Forward 3D FFT** along spatial axes: $\widetilde{\mathbf{P}} = \mathrm{FFT}_{3\mathrm{D}}[\hat{\mathbf{P}}]$.
4. **Pointwise tensor product** in Fourier space:

$$\widetilde{Y}_{i,\ell} = \sum_{j=1}^{3} \widetilde{G}_{ij}\ \widetilde{P}_{j,\ell} \qquad \text{(at each spatial frequency)}, \tag{17}$$

   i.e. a 3×3 matrix times a $3 \times L$ matrix at every grid point.
5. **Inverse 3D FFT**: $\hat{\mathbf{Y}} = \mathrm{FFT}_{3\mathrm{D}}^{-1}[\widetilde{\mathbf{Y}}]$.
6. **Extract** the physical region $\hat{\mathbf{Y}}[{:}N_x, {:}N_y, {:}N_z, :, :]$ and flatten back to $(3N_xN_yN_z, L)$.

The full preconditioned MVP including the diagonal is

$$\mathbf{A}\,\boldsymbol{S} = \mathrm{diag}(\mathbf{A}) \cdot \boldsymbol{S} + \mathbf{A}_{\text{off}}\,\boldsymbol{S}, \tag{18}$$

where $\mathrm{diag}(\mathbf{A}) = \boldsymbol{\alpha}^{-1}$ is applied element-wise and $\mathbf{A}_{\text{off}}$ is evaluated via the FFT procedure above.

3.4 Computational complexity and memory

- **Time**: Each MVP costs $O(N_{\text{cuboid}} \log N_{\text{cuboid}})$ where $N_{\text{cuboid}} = N_xN_yN_z$. For the block variant with L right-hand sides, the L columns are processed in a single batched FFT call.
- **Memory**: The dominant allocations are:
  - $\widetilde{\mathbf{G}}$: $2N_x \times 2N_y \times 2N_z \times 9$ complex128 entries $= 1152\,N_{\text{cuboid}}$ bytes (static, allocated once);
  - padded polarisation arrays (forward and inverse FFT): $2N_x \times 2N_y \times 2N_z \times 3 \times L$ complex128 entries each $= 384\,L\,N_{\text{cuboid}}$ bytes (two such arrays at peak).

  Peak memory $\approx (1152 + 768\,L)\,N_{\text{cuboid}}$ bytes.

3.5 Multi-threaded FFT

The 3D FFTs are computed by `scipy.fft.fftn` with multi-threading. The default number of worker threads is $\max(1,\ n_{\text{cpu}} - 2)$, leaving two cores free for other processes. This can be overridden at runtime via the environment variable `DDA_FFT_WORKERS`.

4 Block-Krylov Iterative Solvers

The DDA system (4) is solved for L right-hand sides simultaneously. Stacking the L incident-field vectors as columns of a block matrix $\mathbf{B}$ (shape $3N \times L$), the problem becomes

$$\mathbf{A}\,\mathbf{X} = \mathbf{B}, \tag{19}$$

where $\mathbf{X}$ (shape $3N \times L$) contains the L solution vectors.

4.1 Jacobi preconditioning

A left Jacobi (diagonal) preconditioner is applied:

$$\mathbf{M}^{-1} = \mathrm{diag}(\mathbf{A})^{-1} = \mathrm{diag}(\alpha_{c,i}), \tag{20}$$

so that the preconditioned system becomes $\mathbf{M}^{-1}\mathbf{A}\,\mathbf{X} = \mathbf{M}^{-1}\mathbf{B}$.

4.2 Block-BiCGSTAB

The primary solver is Block-BiCGSTAB [5], which extends the stabilised bi-conjugate gradient method to block systems. It is suitable for general (non-symmetric) matrices and is the default in `block-DDA_Py`.

The algorithm is summarised in Table 1. Each iteration requires two block MVPs (lines 1 and 4). The default convergence parameters are $\tau = 0.01$ and $n_{\max} = 25$.

Table 1: Block-BiCGSTAB with Jacobi preconditioning (after Tadano et al. [5]).

Input: $\mathbf{A}$, $\mathbf{B}$ (shape $3N \times L$), $\mathbf{M} = \mathrm{diag}(\mathbf{A})$, tolerance τ, max iterations $n_{\max}$.

Initialise: $\mathbf{X}_0 = \mathbf{0}$, $\mathbf{R}_0 = \mathbf{M}^{-1}\mathbf{B}$, $\mathbf{P}_0 = \mathbf{R}_0$, $\widetilde{\mathbf{R}}_0 = \mathbf{R}_0$.

For $n = 0, 1, \ldots, n_{\max} - 1$:

1. $\mathbf{V}_n = \mathbf{M}^{-1}\mathbf{A}\,\mathbf{P}_n$
2. $\boldsymbol{\alpha}_n = (\widetilde{\mathbf{R}}_0^{\mathsf{H}}\,\mathbf{V}_n)^{-1}\,\widetilde{\mathbf{R}}_0^{\mathsf{H}}\,\mathbf{R}_n$
3. $\mathbf{T}_n = \mathbf{R}_n - \mathbf{V}_n\,\boldsymbol{\alpha}_n$
4. $\mathbf{Z}_n = \mathbf{M}^{-1}\mathbf{A}\,\mathbf{T}_n$
5. $\xi_n = \mathrm{tr}(\mathbf{Z}_n^{\mathsf{H}}\,\mathbf{T}_n)\;/\;\mathrm{tr}(\mathbf{Z}_n^{\mathsf{H}}\,\mathbf{Z}_n)$
6. $\mathbf{X}_{n+1} = \mathbf{X}_n + \mathbf{P}_n\,\boldsymbol{\alpha}_n + \xi_n\,\mathbf{T}_n$
7. $\mathbf{R}_{n+1} = \mathbf{T}_n - \xi_n\,\mathbf{Z}_n$
8. If $\|\mathbf{R}_{n+1}\|/\|\mathbf{M}^{-1}\mathbf{B}\| < \tau$: stop, return $\mathbf{X}_{n+1}$.
9. $\boldsymbol{\beta}_n = (\widetilde{\mathbf{R}}_0^{\mathsf{H}}\,\mathbf{V}_n)^{-1}\,(-\widetilde{\mathbf{R}}_0^{\mathsf{H}}\,\mathbf{Z}_n)$
10. $\mathbf{P}_{n+1} = \mathbf{R}_{n+1} + (\mathbf{P}_n - \xi_n\,\mathbf{V}_n)\,\boldsymbol{\beta}_n$

4.3 Block-COCG-rq (alternative solver)

Since the DDA interaction matrix $\mathbf{A}$ is complex symmetric ($\mathbf{A} = \mathbf{A}^{\top}$), a solver specialised for this structure, Block-COCG-rq [6], is also provided. Block-COCG-rq requires only one block MVP per iteration and is therefore faster per iteration than Block-BiCGSTAB. However, for particles with high refractive index it exhibits poorer convergence stability compared to Block-BiCGSTAB. To switch the solver, replace the import and function call in `bl_dda/scatterer.py` (line 3 and line 169): change `bl_bicgstab_jacobi_mvp_fft` to `bl_cocg_rq_jacobi_mvp_fft`. Both functions share the same interface.

5 Computational Performance: Old vs. New Implementation

This section compares the computational cost of the previous implementation (Barrowes 2001 block-Toeplitz FFT with `ray` multiprocessing) and the current implementation (Goodman 1991 3D FFT with NumPy/SciPy broadcasting).

5.1 MVP complexity

Both algorithms compute the off-diagonal matrix–vector product via FFT-based circular convolution, and both have the same asymptotic complexity per right-hand side (RHS):

$$T_{\mathrm{MVP}}^{(1)} = O(N \log N),$$

where $N = N_x N_y N_z$ is the total cuboid grid size. For L right-hand sides the total floating-point operation count is $O(LN \log N)$ in both cases. The difference lies in execution efficiency.

5.2 Parallelisation overhead

Table 2: Execution-efficiency comparison (per Krylov iteration = two block MVPs for Block-BiCGSTAB).

	Old (`ray`)	New (broadcasting)
Parallelism granularity	Process-level (1 MVP / worker)	Thread-level (FFT + BLAS)
Inter-process communication	Array serialise/deserialise	None (single process)
MVP wall time (L RHS)	$\sim \frac{L}{P} T_{\mathrm{MVP}} + T_{\mathrm{ray}}$	$\sim \frac{L}{W} T_{\mathrm{FFT}} \times 2 + T_{\mathrm{matmul}}$
`ray` overhead T_{ray}	$\sim$ 10–100 ms / MVP	0
Worker startup	$\sim$ 1–5 s (pool init)	0
Cache efficiency	Low (each worker holds copy)	High (contiguous memory)

Here P is the number of `ray` workers and W is the number of `scipy.fft` threads (default: `cpu_count` $- 2$).

5.3 Worked example

For $N = 30\,000$, $L = 50$, $P = W = 8$, and 10 Krylov iterations (20 MVPs):

Table 3: Estimated wall-time breakdown.

Cost factor	Old (`ray`)	New (broadcasting)
Effective MVP calls	$20 \times \lceil 50/8 \rceil = 140$	20 (batched)
`ray` IPC overhead	$\sim 140 \times 50\,\mathrm{ms} = 7\,\mathrm{s}$	0
Estimated speedup	1× (baseline)	$\sim$ 3–5×

The three main sources of speedup, in decreasing order of impact, are:

1. **Removal of `ray` IPC overhead.** Array serialisation/deserialisation is eliminated; the effect grows with N.
2. **Batched FFT.** All L RHS are processed in a single `scipy.fft` call; the internal thread pool distributes work efficiently.
3. **Memory locality.** Contiguous-array access within a single process improves CPU cache hit rates.

5.4 Memory comparison

Table 4: Peak memory usage.

	Old (`ray`)	New (Goodman)
Interaction tensor	Copy per worker: $\sim P \times 1152\,N$ bytes	Single copy: $1152\,N$ bytes
Polarisation (during MVP)	1 RHS / worker	All L RHS: $768\,L\,N$ bytes
Total peak	$\sim P \times 1152\,N$	$(1152 + 768\,L) \times N$

For $P = 8$ and moderate L, the old implementation uses roughly $8\times$ more memory for the interaction tensor alone. The new implementation's memory is dominated by the block polarisation arrays when L is large.

6 Scattering Observables

Once the polarisation $\boldsymbol{P}_i$ $(i = 1, \ldots, N)$ has been obtained for each orientation ℓ, the following observables are computed. Throughout this section, k denotes the wavenumber in the medium as defined in (1).

6.1 Absorption and extinction cross sections

The absorption cross section is [2]

$$C_{\mathrm{abs}} = 4\pi k \sum_{i=1}^{N} \mathrm{Im}\big(\boldsymbol{P}_i \cdot \boldsymbol{E}_i^{*}\big), \tag{21}$$

where $\boldsymbol{E}_i$ is the internal electric field at dipole i, reconstructed from the polarisation via

$$\boldsymbol{E}_i = \frac{4\pi\,\boldsymbol{P}_i}{(\epsilon_{r,i} - 1)\,v}. \tag{22}$$

The extinction cross section is

$$C_{\mathrm{ext}} = 4\pi k \sum_{i=1}^{N} \mathrm{Im}\big(\boldsymbol{P}_i \cdot \boldsymbol{E}_{\mathrm{inc},i}^{*}\big). \tag{23}$$

The corresponding efficiency factors are $Q_{\mathrm{abs}} = C_{\mathrm{abs}}/(\pi r_{ve}^2)$ and $Q_{\mathrm{ext}} = C_{\mathrm{ext}}/(\pi r_{ve}^2)$, where r_{ve} is the volume-equivalent sphere radius:

$$r_{ve} = \left(\frac{3\,N\,v}{4\pi}\right)^{1/3}. \tag{24}$$

6.2 Forward scattering amplitude (PCAS)

The forward scattering amplitude in the PCAS (Polarimetric Complex Amplitude of Scattering) framework is computed from the far-field projection of the dipole polarisations.

For the forward scattering direction $\hat{\boldsymbol{u}}_{\mathrm{fw}} = +\hat{\boldsymbol{z}}$ in the laboratory frame, define the phase factor for each dipole:

$$e_{\mathrm{fw},i} = \exp\big(ik\,\hat{\boldsymbol{u}}_{\mathrm{fw}} \cdot \boldsymbol{r}_i\big). \tag{25}$$

The θ- and ϕ-components of the forward scattering amplitude correspond to the s- and p-polarisation channels of the CAS-v2 detector, respectively: $S_{\mathrm{fw},\theta}^{\mathrm{PCAS}} \equiv S_s(0)$ and $S_{\mathrm{fw},\phi}^{\mathrm{PCAS}} \equiv S_p(0)$. They are computed as

$$S_{\mathrm{fw},\theta}^{\mathrm{PCAS}}\ [= S_s(0)] = k^2 \sum_{i=1}^{N} (\boldsymbol{P}_i \cdot \hat{\boldsymbol{e}}_{\theta,\mathrm{fw}})\ (\sqrt{2}\, e_{\mathrm{fw},i})^*, \tag{26}$$

$$S_{\mathrm{fw},\phi}^{\mathrm{PCAS}}\ [= S_p(0)] = k^2 \sum_{i=1}^{N} (\boldsymbol{P}_i \cdot \hat{\boldsymbol{e}}_{\phi,\mathrm{fw}})\ (\sqrt{2}\, i\, e_{\mathrm{fw},i})^*, \tag{27}$$

where the factors $\sqrt{2}$ and $\sqrt{2}\,i$ arise from the circular-polarisation decomposition of the incident field (7).

6.3 Backward scattering amplitude (OCBS)

For the exact backward scattering direction $\hat{\boldsymbol{u}}_{\mathrm{bk}} = -\hat{\boldsymbol{z}}$, define the phase factor

$$e_{\mathrm{bk},i} = \exp\!\big(ik\, \hat{\boldsymbol{u}}_{\mathrm{bk}} \cdot \boldsymbol{r}_i\big). \tag{28}$$

Let $\hat{\boldsymbol{e}}_{\theta,\mathrm{bk}}$ and $\hat{\boldsymbol{e}}_{\phi,\mathrm{bk}}$ be the polar and azimuthal unit vectors associated with the backward direction. In the laboratory frame: $\hat{\boldsymbol{e}}_{\theta,\mathrm{bk}} = -\hat{\boldsymbol{x}}$ and $\hat{\boldsymbol{e}}_{\phi,\mathrm{bk}} = +\hat{\boldsymbol{y}}$.

The θ- and ϕ-components of the backward scattering amplitude are

$$S_{\mathrm{bk},\theta}^{\mathrm{OCBS}} = k^2 \sum_{i=1}^{N} (\boldsymbol{P}_i \cdot \hat{\boldsymbol{e}}_{\theta,\mathrm{bk}})\ (\sqrt{2}\, e_{\mathrm{bk},i})^*, \tag{29}$$

$$S_{\mathrm{bk},\phi}^{\mathrm{OCBS}} = k^2 \sum_{i=1}^{N} (\boldsymbol{P}_i \cdot \hat{\boldsymbol{e}}_{\phi,\mathrm{bk}})\ (\sqrt{2}\, i\, e_{\mathrm{bk},i})^*, \tag{30}$$

where the reference-wave factors $\sqrt{2}$ and $\sqrt{2}\,i$ are the same as in the forward case.

The OCBS (Opposite-Circular-Basis Scattering) amplitude is then

$$S_{\mathrm{bk}}^{\mathrm{OCBS}} = \frac{-S_{\mathrm{bk},\theta}^{\mathrm{OCBS}} + S_{\mathrm{bk},\phi}^{\mathrm{OCBS}}}{\sqrt{2}}. \tag{31}$$

More precisely, the θ- and ϕ-components correspond to combinations of the Mishchenko amplitude matrix elements [9]:

$$S_{\mathrm{bk},\theta}^{\mathrm{OCBS}} \equiv S_{11}(\pi) + i\, S_{12}(\pi), \qquad S_{\mathrm{bk},\phi}^{\mathrm{OCBS}} \equiv S_{22}(\pi) - i\, S_{21}(\pi). \tag{32}$$

The backscattering theorem $S_{21}(\pi) = -S_{12}(\pi)$ causes the off-diagonal contributions to cancel when the two components are combined in (31), yielding an expression equivalent to Eq. (28) of Moteki and Adachi [10].

7 Gaussian Random Ellipsoid (GRE) Shape Model

The particle shape is generated using the Gaussian Random Ellipsoid (GRE) model of Muinonen & Pieniluoma [7].

7.1 Base ellipsoid

The base ellipsoid is defined by three semi-axes $a \geq b \geq c$ and the volume-equivalent radius r_v. Two axis ratios are specified as input: b/c (`bc_ratio`) and a/b (`ab_ratio`). The semi-axis c is determined by volume conservation:

$$c = \left(\frac{r_v^3}{(a/b)\,(b/c)^2} \right)^{1/3}, \qquad b = c \cdot (b/c), \qquad a = b \cdot (a/b). \tag{33}$$

7.2 Surface deformation

A Gaussian random field $s(\theta, \phi)$ is sampled on the surface of the base ellipsoid with covariance

$$\mathrm{Cov}(s_i, s_j) = \beta^2 \exp\left(-\frac{\|\boldsymbol{r}_i - \boldsymbol{r}_j\|^2}{2\,\ell_c^2}\right), \tag{34}$$

where β is the standard deviation of the deformation and ℓ_c is the correlation length, which controls the fineness of the surface roughness. In the current version (v0.6.0), $\ell_c = 0.3\,c$ is adopted as an empirical choice.

The deformed surface position is obtained by displacing each point on the base ellipsoid along the outward surface normal $\hat{\boldsymbol{n}}_0$:

$$\boldsymbol{r}_{\mathrm{GRE}}(\theta, \phi) = \boldsymbol{r}_{\mathrm{base}}(\theta, \phi) + h_0\big[\exp\big(s(\theta, \phi)\big) - \tfrac{1}{2}\beta^2 - 1\big]\ \hat{\boldsymbol{n}}_0(\theta, \phi), \tag{35}$$

where $h_0 = c^2/a$ is a length scale and $\hat{\boldsymbol{n}}_0$ is the unit outward normal to the base ellipsoid.

7.3 Lattice generation and interior detection

The lattice spacing is determined by the number of dipoles per wavelength inside the particle (dpl):

$$d = \frac{\lambda_{\mathrm{particle,min}}}{\mathrm{dpl}} = \frac{\lambda_0}{\max_c |m_{p,c}| \times \mathrm{dpl}}, \tag{36}$$

where $m_{p,c}$ ($c \in \{x, y, z\}$) is the complex particle refractive index along axis c (cf. Sec. 2.4) and $\lambda_{\mathrm{particle,min}} = \lambda_0 / \max_c |m_{p,c}|$ is the shortest wavelength inside the particle. The default value is $\mathrm{dpl} = 17$, which is a typical choice for DDA simulations of sub-micrometre particles. A cubic lattice with this spacing is generated within the bounding box of the deformed surface. Interior points are identified by a three-axis ray-casting algorithm.

7.4 Volume-preserving lattice rescaling

Because the GRE surface is discretised onto a cubic lattice, the total volume of the N occupied dipoles, Nd^3, generally differs from the target volume $V_{\mathrm{target}} = \frac{4}{3}\pi r_{v,\mathrm{base}}^3$ by a few percent. When parameter sweeps require exact control of the volume-equivalent radius (e.g. an equally-spaced $r_{v,\mathrm{base}}$ grid), this discretisation error is undesirable.

To eliminate it, the lattice spacing is rescaled *after* the interior detection step. Given the number of occupied sites N and the desired $r_{v,\mathrm{base}}$, the adjusted spacing is

$$d_{\mathrm{adj}} = \left(\frac{V_{\mathrm{target}}}{N}\right)^{1/3} = \left(\frac{4\pi}{3N}\right)^{1/3} r_{v,\mathrm{base}}. \tag{37}$$

All lattice coordinates are then multiplied by the scale factor $s = d_{\mathrm{adj}}/d$, and d_{adj} replaces d in every subsequent computation: the Green's function (8), the Clausius–Mossotti polarisability (10), and the radiative correction (13). Because both the inter-dipole distances and the dipole volume $v = d_{\mathrm{adj}}^3$ change by the same factor, the DDA formulation remains self-consistent.

This rescaling is performed in the `Target` constructor (`bl_dda/scatterer.py`), which requires `r_v_base` as a mandatory argument.

8 Spheroid Symmetry and Orientation Averaging

For spheroids (axially symmetric particles with $a = b$ and GRE roughness $\beta_{\mathrm{GRE}} = 0$), the scattering amplitudes possess analytical constraint relations that enable significant computational savings in multi-orientation calculations.

8.1 Euler angle mapping

The intrinsic ZYZ Euler angles (α, β, γ) relate to the spheroid geometry as follows:

- β: polar angle of the symmetry axis (tilt from the laboratory z-axis),
- α: azimuthal angle of the symmetry axis (rotation of the tilt plane about z),
- γ: rotation about the symmetry axis; irrelevant for a spheroid ($a = b$).

In the notation of Sec. 8.2, $\theta \equiv \beta$ and $\phi \equiv \alpha$.

8.2 Constraint relations at forward scattering

When the symmetry axis lies in the xz-plane ($\alpha = 0$), the cross-polarisation elements of the amplitude scattering matrix vanish: $S_{12}(\theta, 0) = S_{21}(\theta, 0) = 0$. The forward-scattering amplitudes at arbitrary α are then determined entirely by the two diagonal elements at $\alpha = 0$:

$$S_s(\theta, \alpha) = \frac{S_{11}(\theta,0) + S_{22}(\theta,0)}{2} + \frac{S_{11}(\theta,0) - S_{22}(\theta,0)}{2}\, e^{i2\alpha}, \tag{38}$$

$$S_p(\theta, \alpha) = \frac{S_{11}(\theta,0) + S_{22}(\theta,0)}{2} - \frac{S_{11}(\theta,0) - S_{22}(\theta,0)}{2}\, e^{i2\alpha}, \tag{39}$$

where $S_s \equiv S_{11} + iS_{12}$ and $S_p \equiv S_{22} - iS_{21}$ are the PCAS observables (Sec. 6.2).

These imply three verifiable symmetries:

1. **Trace invariance:** $S_s(\alpha) + S_p(\alpha) = S_{11}(\theta,0) + S_{22}(\theta,0) = \text{const}$.
2. **Modulus preservation:** $|S_s(\alpha) - S_p(\alpha)| = |S_{11}(\theta,0) - S_{22}(\theta,0)| = \text{const}$.
3. **Cross-symmetry:** $S_s(\theta,\, \alpha + \pi/2) = S_p(\theta,\, \alpha)$.

These relations are verified numerically at machine precision ($\sim 10^{-16}$) in `test_spheroid_symmetry.ipynb`.

8.3 Analytical phi-averaging (labor-saving mode)

For orientation-averaged observables, the azimuthal average over $\alpha \in [0, 2\pi]$ eliminates the oscillatory term $e^{i2\alpha}$, yielding

$$\langle S_s(\theta)\rangle_\alpha = \langle S_p(\theta)\rangle_\alpha = \frac{S_{11}(\theta,0) + S_{22}(\theta,0)}{2}. \tag{40}$$

Furthermore, C_{ext} and C_{abs} are also α-invariant for a spheroid, so the values computed at $\alpha = 0$ are identical to their α-averages.

Implementation. When `run_dda.py` detects a spheroid (`ab_ratio == 1` and `gre_beta == 0`), it activates *spheroid mode*:

- The DDA is solved only for N_β orientations at $\alpha = 0$, $\gamma = 0$, with $\cos\beta$ at the N_β equal-area midpoints (Sec. 2.2).
- The full $N_\alpha \times N_\beta \times N_\gamma$ orientation grid is then filled analytically using the constraint relations (38)–(39):

  $$S_s(\alpha_j, \beta_i) = A_i + B_i\, e^{i2\alpha_j}, \qquad S_p(\alpha_j, \beta_i) = A_i - B_i\, e^{i2\alpha_j},$$

  where $A_i = [S_s(\beta_i, 0) + S_p(\beta_i, 0)]/2$ and $B_i = [S_s(\beta_i, 0) - S_p(\beta_i, 0)]/2$.
- For backward scattering, the OCBS amplitude follows the same rotational structure: $S_{\text{bk}}^{\text{OCBS}}(\alpha_j, \beta_i) = S_{\text{bk}}^{\text{OCBS}}(0, \beta_i)\, e^{i2\alpha_j}$.
- C_{ext} and C_{abs} are α-invariant, so the $\alpha = 0$ values are broadcast to all grid points.

This reduces the number of DDA solves from $N_\alpha \times N_\beta \times N_\gamma$ to N_β, while the output grid retains the full $N_\alpha \times N_\beta \times N_\gamma$ resolution with *exact* (not interpolated) values at every grid point.

Applicability. These relations hold for any particle whose cross-section in the plane perpendicular to the symmetry axis is circular ($a = b$), regardless of the axial ratio b/c. They break down for particles lacking a continuous rotational symmetry axis (e.g., tri-axial ellipsoids with $a \neq b$, or particles with GRE roughness $\beta_{\mathrm{GRE}} > 0$).

Motivation. For spheroids, the T-matrix method [9] is generally far more efficient than DDA. However, the extended-boundary-condition method (EBCM) used to compute the T-matrix becomes numerically unstable for extreme aspect ratios ($b/c \ll 1$ or $b/c \gg 1$), where the surface integrals over highly elongated or flattened geometries lead to severe loss of numerical precision [9]. In such regimes, DDA with the spheroid-mode optimisation described above provides a viable alternative: the computation remains stable regardless of the aspect ratio, and the analytical α-averaging reduces the cost by a factor of N_α relative to the general DDA mode.

9 Mie Theory Reference

For validation, the DDA solution for a volume-equivalent sphere is compared to the exact Mie theory [8].

9.1 Partial-wave expansion

For a homogeneous sphere of radius r_p, the size parameter is $x = k\, r_p = 2\pi\, m_m\, r_p/\lambda_0$ and the relative refractive index is $m_r = m_p/m_m$. The number of partial-wave terms is

$$n_{\max} = \lfloor |x + 4\, x^{1/3} + 2| \rfloor. \tag{41}$$

The Mie coefficients a_n and b_n are computed from Riccati–Bessel functions ψ_n, χ_n, $\xi_n = \psi_n + i\chi_n$ and the logarithmic derivative $D_n(m_r x)$ via downward recurrence [8].

9.2 Efficiencies and scattering amplitudes

$$Q_{\mathrm{ext}} = \frac{2}{x^2} \sum_{n=1}^{n_{\max}} (2n+1)\ \mathrm{Re}(a_n + b_n), \tag{42}$$

$$Q_{\mathrm{sca}} = \frac{2}{x^2} \sum_{n=1}^{n_{\max}} (2n+1)\, (|a_n|^2 + |b_n|^2), \tag{43}$$

$$Q_{\mathrm{abs}} = Q_{\mathrm{ext}} - Q_{\mathrm{sca}}. \tag{44}$$

The angular scattering amplitudes are [8]

$$S_1(\theta) = \sum_{n=1}^{n_{\max}} \frac{2n+1}{n(n+1)} \big(a_n\, \pi_n(\cos\theta) + b_n\, \tau_n(\cos\theta)\big), \tag{45}$$

$$S_2(\theta) = \sum_{n=1}^{n_{\max}} \frac{2n+1}{n(n+1)} \big(a_n\, \tau_n(\cos\theta) + b_n\, \pi_n(\cos\theta)\big), \tag{46}$$

where π_n and τ_n are the angular functions computed via upward recurrence. The forward and backward scattering amplitudes used for validation are obtained from S_1 and S_2 at $\theta = 0$ and $\theta = \pi$.

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